

Latent Undertow: How Ordinary Typos Break Probes

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Abstract

LLMs handle ordinary typing variation fluently: a typo or missing punctuation leaves both user intent and the model’s response substantively unchanged. Yet probes that detect malicious prompts by reading the model’s hidden states tell a different story: the same edit rotates the readout vector by 43° – 56° at the perturbed token, decaying below 15% within ≈ 10 downstream tokens. Stacking ≈ 3 common typos per message cuts a single-position prompt-injection probe’s TPR@FPR=1% by 12.0pp, a gap recalibration alone cannot close. Multi-position aggregation cures localized perturbations (≤ 0.5 pp loss) but only attenuates distributed ones, where even attention- and max-based aggregators still drop ~ 3.8 pp. For single-position probes, we introduce a KV-cache fork: a short fixed suffix appended after the user message lets the probe read a few tokens downstream of the perturbation, exploiting its rapid spatial decay. This closes 95% of the gap (-0.6 pp residual)—an order of magnitude better than perturbation-augmented training (-3.7 pp). The rotation-and-decay geometry replicates on Llama-3.1-8B, Qwen3-8B, and Gemma-4-E4B; probe evaluation is on Llama-3.1-8B. Code: <https://github.com/eladd-ai/latent-undertow>.

ployment, however, requires reliability under the varied, imperfect text that users actually send: over 40% of real user inputs to LLM systems contain typographical errors or grammatical noise (Wang et al., 2024).

The mismatch is structured. The activation perturbation is sharp but local: a large rotation at the perturbed token, decaying within a handful of downstream positions (§5.2, Figure 2). The same geometry that breaks single-position probes—an isolated burst at one token—is what makes the failure repairable: a readout placed a few tokens downstream of the perturbation reads activations that have already settled. We call this the *latent undertow*: brief, localized internal movement under fluent surface behavior.

We make three contributions:

1. We characterize the spatial structure of the activation shift: a large localized rotation at the perturbed token that decays rapidly downstream.
2. Stacking common typing errors as a stress test reduces single-position probe TPR by 12.0pp at FPR=1%, not recoverable by threshold recalibration alone.
3. We evaluate remedies along two axes. Multi-position aggregation cures localized perturbations (≤ 0.5 pp loss) but only attenuates distributed ones, where even attention- and max-based aggregators still drop ~ 3.8 pp (§6.2). For single-position probes, a KV-cache-forked probe suffix relocates the readout downstream of the perturbation, closing 95% of the gap (-0.6 pp residual)—an order of magnitude better than perturbation-augmented training (-3.7 pp) (§7).

1. Introduction

Activation-based probes have become a practical tool for monitoring LLM behavior at inference time. Linear probes trained on hidden-state activations recover task-relevant properties of model internals (Alain & Bengio, 2017), detect unsafe user intent (Zou et al., 2025), and flag prompt injection attempts (Abdelnabi et al., 2025). Production de-

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2. Background and Related Work

Activation-based probing. Linear probes on intermediate representations have long been used to recover task-relevant properties from deep network representations (Alain & Bengio, 2017); applied to LLMs, probes trained on the hidden state at the *final token position* detect latent model state (Azaria & Mitchell, 2023; Marks & Tegmark, 2024) and unsafe behavior (Zou et al., 2025). Production deployments adopt the same single-position approach to detect prompt injection attacks (Abdelnabi et al., 2025; Zou et al., 2026) and other safety violations. Kramár et al. (2026)

extend this to multi-position aggregation architectures for production Gemini probes, building on a longer lineage of pooling-based extensions (max-pool, attention pool) over token sequences in transformer classification (Behrendt et al., 2025); we adopt their taxonomy as our evaluation framework (§6.2).

Output-level robustness. LLM outputs are sensitive to surface prompt variation across paraphrase, format, and template choice (Salinas & Morstatter, 2024; Sclar et al., 2024; Mizrahi et al., 2024), including typing-specific perturbations (Gan et al., 2024). Output sensitivity implies that activations shift as well. Yet under non-adversarial typing noise the LLM’s reading of user intent is largely preserved (Aliakbarzadeh et al., 2024; Zhao et al., 2026) (§3.3). This raises the question we investigate: when surface variation perturbs activations without altering the model’s reading of intent, what do activation-based probes actually read out?

Robustness of probes. Whether surface input variation affects the internal representations that probes read from has received less attention than output-level sensitivity. The closest prior work (Yu et al., 2026) applies *targeted* character-level perturbations to GPT-4o-mini-selected harmful keywords on AdvBench and reports diagnostic-probe accuracy dropping from ~95% to ~80% on four open-source models (Llama-3-8B, Mistral-7B, Vicuna-7B/13B). We extend this line to *ordinary, non-targeted typing variation*, without adversarial intent.

Post-user prompt-engineering defenses. Sandwich defenses append validation/reminder text after user inputs to resist prompt injection in the *behavioral* pipeline (Liu et al., 2024). We show that a similar post-user mechanism dilutes perturbation signal in the *activation* pipeline while preserving probe discriminability, exploited through KV-cache forking to build a probe path that is user-transparent and robust.

3. Framework

3.1. Definitions

We study *surface perturbations*: character-level edits that preserve user intent and (typically) the LLM’s response. The probe target throughout is binary user intent: malicious (prompt injection or unsafe request) versus benign.

A perturbation is *intent-preserving* if a human reader interprets the perturbed and clean inputs as expressing the same user intent, and *behavior-preserving* if the LLM’s response is unchanged. Probes target intent, so intent-preservation is the criterion our fragility claims require. The perturbations we study are intent-preserving by construction; behavior preservation is not guaranteed under heavy perturbation

(§3.3 gives an example).

We further distinguish two perturbation regimes by spatial structure: *localized* (clustered near the readout) and *distributed* (spread across the input).

3.2. Perturbation Taxonomy

We study the following surface-perturbation families, individually and in combination:

- **Adjacent-key typo:** QWERTY neighbor substitution on last word or scattered mid-message word.
- **Punctuation:** trailing period toggled; question mark → period.
- **Capitalization:** shift-too-early (letter uppercased before punctuation); first letter decapitalization.
- **Omission:** missing space after punctuation.
- **Stacked-typing bundle:** a localized stress-test composition of the above primitives, averaging 3.3 character edits per sample (defined in §6).

The first four families are individual primitives; the stacked-typing bundle is the localized stress test introduced in §6. For the distributed regime we use **every-second-word adjacent-key typos** (~50% of content tokens corrupted; used as a stress test rather than a model of typical user noise; §6.2); the type survey in §5.3 additionally evaluates **question→/** as a severe single-perturbation case.

Adjacent-key typos serve as the primary vehicle for mechanistic characterization (§5–§5.3) because they occur at arbitrary positions, enabling clean isolation of spatial decay independent of message position. Terminal punctuation substitutions (e.g., question→/) produce larger absolute rotations but are structurally constrained to the message end; their magnitudes appear in Table 1, and Appendix C verifies the decay shape generalizes.

3.3. Behavioral Invariance Premise

The probe experiments below assume surface perturbations are intent-preserving (§3.1). We illustrate this on four clean-vs-perturbed examples spanning both perturbation regimes and label classes (Llama-3.1-8B-Instruct; full prompts in Appendix A), and verify it more broadly with an LLM-as-judge evaluation across all three models that confirms intent preservation in 94.6% of pairs (Appendix A.2). In the **localized** regime the LLM’s response is functionally unchanged on both benign and malicious cases; the probe is the sole failure point. In the **distributed** regime intent is preserved but the malicious example bypasses the LLM’s safety alignment—a non-adversarial demonstration of alignment brittleness under heavy surface noise, paralleling Gu et al. (2025)’s finding that small adversarial shifts in hidden activations can re-trigger unsafe outputs in aligned mod-

els. LLM-judge response equivalence is high under the bundle regime and lower under ESW, consistent with the regime’s known alignment brittleness. Together these justify the premise; §6–§7 measure the activation- and probe-level consequences.

4. Experimental Setup

4.1. Models

We evaluate three instruction-tuned models from different training lineages and release generations: **Llama-3.1-8B-Instruct**, **Qwen3-8B**, and **Gemma-4-E4B**. The cross-model design verifies that the activation-level mechanism characterized in §5–§5.2 (large on-site rotation, rapid spatial decay) transfers across families: all three reproduce the mechanism qualitatively, with quantitative magnitudes differing between families but the structural pattern consistent. Activations are extracted at the last token of the user turn, consistent with standard single-position probing (Azaria & Mitchell, 2023; Zou et al., 2025), via `HuggingFace transformers` with `sdpa` attention.

4.2. Activation-Level Experiments

Activation characterization (§5–§5.2) uses four depth checkpoints per model (Llama-3.1-8B: 8/16/24/31; Qwen3-8B: 9/18/27/35; Gemma-4-E4B: 28/29/40/41, contrasting sliding-window and full-attention layers). Each condition uses 100 OpenOrca prompts (100–300 words), with adjacent-key typos applied at up to four positions per prompt. Effect sizes (on-site rotations of 43–56°) are large relative to per-prompt variance, so structural comparisons are statistically robust at this scale.

4.3. Probe-Level Experiments

Probe-consequence and multi-architecture experiments (§6–§7) use Llama-3.1-8B layer 31 (final residual stream, last user token); late Llama layers are competitive for intent classification under distribution shift (Fomin, 2026), and we reproduce the layer sweep on our corpus (Appendix G). Cross-model probe evaluation is left to follow-up: the mechanism transfer above suggests qualitative similarity, but probe-level quantitative outcomes also depend on data and training.

Classification task. Probes evaluate binary classification of user intent: benign requests (instruction-following, coding, customer-support, creative writing) versus malicious inputs (prompt injections, jailbreak attempts, unsafe requests, scam messages). This framing reflects a realistic deployment scenario where a probe monitors the user turn before the model responds.

Dataset corpus. We use a curated mix of 29 publicly available datasets (full list in Appendix F). The benign side covers diverse legitimate-use domains (instruction-following, coding, customer support, creative writing, tool use); the malicious side covers five attack families (direct/indirect prompt injection, jailbreak, harmful requests, scam). We sample from each source to keep the training corpus tractable ($N=168,440$) and prevent any dataset from dominating.

Training and evaluation. Unless modified by a defense method (§7), probes train on clean activations at the user-EOT readout; perturbations are applied at inference time. We evaluate under two complementary protocols: 5-fold stratified cross-validation (in-distribution; clean AUC $99.80\% \pm 0.01\%$ across folds) and Leave-One-Dataset-Out following Fomin (2026) (out-of-distribution).

5. Perturbation Effects on LLM Activations

Reporting convention: angular distance. The activation shift is purely directional (norm approximately unchanged, §5), so we describe it by the rotation between x and the perturbed x' . Within this directional framing, we report degrees rather than cosine similarity because the score change of a linear probe with normal w is bounded by the chord length, $\|x' - x\| \approx 2R \sin(\theta/2)$ where $R = \|x\|$, and cosine similarity squashes chord through a squared map. As a concrete example, 10° vs 30° rotations differ in chord by $3\times$ (the relevant ratio for probe score change) but in cosine similarity by only 0.985 vs 0.866. The between-prompt baseline of 34.6° (Table 1) provides a natural calibration scale. Cosine equivalents are given at first occurrence.

5.1. On-Site Impact

A single surface perturbation at token position t produces a large angular rotation of the hidden-state activation at that position (Figure 1). Mean on-site angular change ranges from 43° to 56° (cosine similarity 0.73 to 0.56) across all three models and layers, with within-model standard deviations of 10°–24° (shaded bands) reflecting variation across typo positions and prompts. No systematic trend with depth is observed. Crucially, the *norm* is approximately unchanged: the perturbation is purely directional, ruling out a magnitude confound and implying that the signal is carried in the direction of the representation, not its scale.

Note on tokenization. On-site measurements use adjacent-key perturbations that preserve the BPE token count of the affected word, so the rotation reports the model’s response to a single-token swap rather than to a re-segmentation of the word. Perturbations that re-segment the word (e.g., missing-space, letter insertion) are not mea-

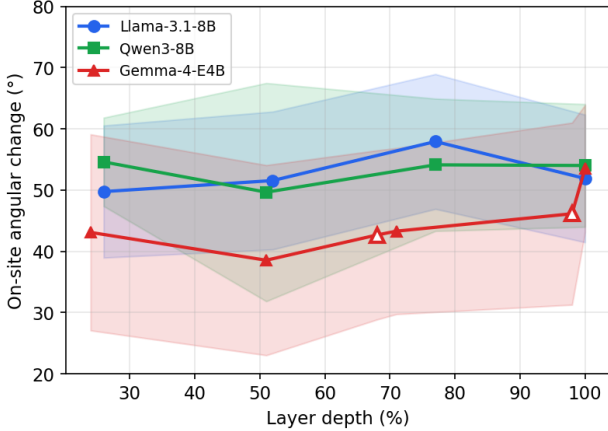


Figure 1. **On-site rotation is large and consistent across models and layers.** Mean on-site angular change at each depth checkpoint (43° – 56° , no systematic depth trend); adjacent-key typos on 100 OpenOrca prompts. Shaded bands: ± 1 std. Gemma-4-E4B: open markers = sliding-window layers, filled = full-attention.

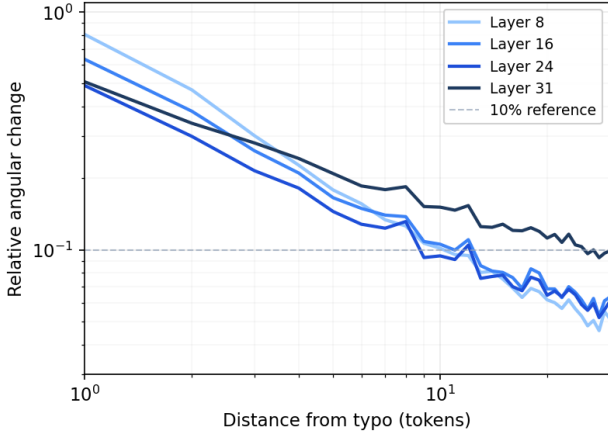


Figure 2. **Spatial decay of perturbation effects across layers.** Relative angular change downstream of the typo site (normalized to 1 at the typo); signal falls below 15% within $d \approx 10$ tokens. Llama-3.1-8B-Instruct, averaged over typo-position conditions.

sured in this section.

5.2. Spatial Structure of the Perturbation Effect

A typo’s on-site rotation (§5) does not propagate freely. Figure 2 shows the relative angular change at positions $d=1, 2, \dots, 30$ tokens downstream of the typo, normalized to 1 at the typo site, averaged over three prompt-length conditions. All four depth checkpoints fall below 15% of the on-site signal by $d \approx 10$ tokens and continue to decay slowly thereafter; the decay profile is consistent across perturbation positions and replicates in Qwen3-8B and Gemma-4-E4B (Appendix B, Figures 6–7).

The forward decay establishes spatial locality: the pertur-

Table 1. **EOT angular change by perturbation type.** Llama-3.1-8B-Instruct, layer 31. d = token distance from typo to EOT readout; % baseline = ratio to the between-prompt angular distance.

Perturbation	d	EOT angle	% baseline
Adjacent-key	10	4.7°	14%
Missing space	10	3.4°	10%
Question $\rightarrow$ period	1	7.9°	23%
Question $\rightarrow$ /	1	26.4°	76%
Between-prompt baseline	—	34.6°	100%

bation effect is concentrated near the typo and rapidly attenuated downstream. Single-position readouts at the perturbation site are exposed; multi-position aggregators see dilution-attenuated signal. §6 examines the consequences; §5.3–§6 focus on Llama-3.1-8B-Instruct.

5.3. Perturbation Type Survey

The spatial decay framework predicts that perturbation impact scales with proximity to the readout position. Table 1 tests this prediction across four perturbation families spanning both mid-sequence and terminal placements. The prediction is confirmed: mid-sequence typos measured at $d=10$ produce modest EOT angles ($\approx 3^\circ$ – 5° , or 10–14% of the between-prompt baseline), while terminal substitutions at $d=1$ bypass spatial attenuation entirely. The most aggressive case (question $\rightarrow$ /) reaches 26.4° , or 76% of the between-prompt baseline, making it the most operationally dangerous single perturbation we tested.

Decay shape is invariant to perturbation severity. Two terminal-punctuation perturbations in Table 1 differ by $\sim 2.4\times$ in on-site magnitude ($? \rightarrow .$ at 24.7° , $? \rightarrow /$ at 59.9°); their relative-decay curves overlap closely (Appendix C, Figure 8).

Co-occurring perturbations. Table 1 characterizes each type in isolation. Figure 3 shows what happens when two typos co-occur, using a controlled design (same prompt, same upstream start position): combined signal is compared to the single-typo-A baseline across three inter-typo token distances. For closely spaced typos ($d_{\text{inter}} \approx 2$ tokens), the combined curve is elevated from the first offset onward ($\approx 1.6\times$ the baseline), as both typos’ spatial footprints overlap immediately. For medium ($d_{\text{inter}} \approx 15$) and separated ($d_{\text{inter}} \approx 25$) conditions, the combined signal tracks the single-typo curve until reaching typo-B’s position, where it spikes before settling to a new elevated level (≈ 1.8 – $2.0\times$). This locality follows directly from forward spatial decay (§5.2): before typo-B, only typo-A contributes; after typo-B, both footprints accumulate at each downstream position.

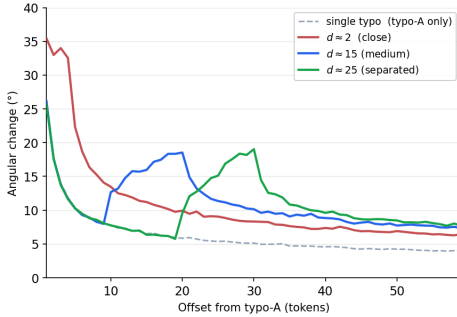


Figure 3. **Two co-occurring typos: combined signal from typo-A forward.** Combined curve (solid) vs. single-typo-A baseline (gray dashed), for three inter-typo distances; dotted verticals mark typo-B’s mean position. Llama-3.1-8B layer 31.

Designing the stacked-typing bundle. This survey motivates the perturbation evaluated in §6. It covers both ends of the severity spectrum: a terminal substitution (question→period, 23% of baseline) alongside two adjacent-key mid-sequence typos and two capitalization/punctuation variants. The most severe single perturbation (question→/) is excluded as an unusually disruptive character choice; the milder terminal variant sits within the range of edits a rushed user actually produces. Restricting to only mid-sequence typos would understate operational fragility; restricting to only terminal substitutions would overstate it.

6. Consequence for Activation-Based Probes

6.1. Single-Position Probe Fragility

We evaluate the operational consequence of the perturbation characterization above using a linear probe trained on clean activations. The probe is a linear classifier trained on clean activations from the 29-dataset corpus by full-batch gradient descent (Adam, weight decay 3×10^{-3} , balanced class weights, standard scaling; full hyperparameters in Appendix G); results are means over 5-fold stratified CV ($\approx 134,750$ train / 33,690 test per fold).

Stacked-typing bundle. We apply a stacked-typing bundle per test sample, combining: (1) trailing question mark → period, (2) adjacent-key QWERTY typo on the last alphabetic word, (3) adjacent-key typo on a word ≥ 3 positions before the last word, (4) shift-too-early capitalization (letter uppercased before punctuation), (5) trailing period toggle. Mean: 3.33 character edits per sample (range 1–5; 99.996% valid coverage). Each component is applied only when its structural prerequisite is met (e.g., terminal punctuation for (1) and (5); alphabetic words at specific distances from the end-of-message token for (2) and (3)), so the number of applied components varies across prompts and averages to 3.33 over the corpus. Each individual component is the kind of edit a real user produces—typos, capitalization errors,

Table 2. **Probe fragility under the stacked-typing bundle.** Linear probe on Llama-3.1-8B-Instruct, layer 31, user-EOT position. “Clean” = evaluation on unperturbed test set; “Perturbed” = same probe on the stacked-typing bundle. Full ROC in Figure 4.

Condition	AUC	TPR@FPR=1%	TPR@FPR=5%
Clean	0.9980	97.4%	99.5%
Perturbed	0.9925	85.4%	96.7%
Δ	−0.55pp	−12.0pp	−2.8pp

punctuation drift—at a rate within the range introduced by rushed or mobile-keyboard typing. We do not claim that our specific 5-component combination matches any measured user distribution; the same pattern, applied deliberately, is also a no-cost attack vector requiring no model access or optimization.

Results. Table 2 shows the probe performance under the stacked-typing bundle. At the deployment-realistic operating point of FPR=1%, TPR drops from 97.4% to 85.4%, a loss of 12.0 percentage points. The AUC drop is small (−0.55pp), so rank order is mostly preserved; but the score distributions shift class-asymmetrically—malicious samples cross below the fixed-FPR threshold more readily than benign samples cross above it, weighting the failure toward false negatives. Threshold recalibration alone cannot recover the clean (FPR, TPR) operating point (Figure 4). Because in-distribution evaluation (5-fold CV) can overstate operational reliability (Fomin, 2026), we additionally evaluate under Leave-One-Dataset-Out (LODO): AUC drops from 0.998 to 0.81–0.92 and accuracy from $> 99\%$ to 0.77–0.81 across the five architectures *before* any perturbation is applied, with large cross-fold variance dominated by which dataset is held out (Appendix I).

6.2. Multi-Architecture Comparison

We compare five probe architectures spanning single-position to full-sequence readout, under both localized and distributed perturbation regimes.

Architectures evaluated. We adopt the probe taxonomy of Kramár et al. (2026) across three readout scopes (full hyperparameters in Appendix G):

- **Single-position:** *Linear (user EOT)*, a linear classifier on the user-EOT activation.
- **Local window:** *Mean Linear (last 16)*, a uniform mean over the last 16 token activations, then linear classifier.
- **Full sequence:** *MLP, Attention, MultiMax*, per-token feature transform aggregated across the user-turn sequence by mean, softmax-attention pooling, or per-head max selection respectively.

Table 3. Multi-architecture robustness under localized (stacked-typing bundle) and distributed (every-second-word) perturbations. Llama-3.1-8B-Instruct, layer 31, 5-fold CV (mean $\pm$ std across folds). Both columns report $\Delta\text{TPR@FPR}=1\%$ (perturbed – clean) so the two regimes are directly comparable.

Probe	$\Delta\text{TPR@FPR}=1\%$	
	Localized (bundle)	Distributed (ESW)
Linear (user EOT)	$-12.00 \pm 0.48\text{pp}$	$-6.85 \pm 0.56\text{pp}$
Mean last 16	$-0.87 \pm 0.09\text{pp}$	$-4.53 \pm 0.33\text{pp}$
MLP (all)	$-0.30 \pm 0.07\text{pp}$	$-16.81 \pm 1.26\text{pp}$
Attention (all)	$-0.48 \pm 0.15\text{pp}$	$-3.78 \pm 1.76\text{pp}$
MultiMax (all)	$-0.48 \pm 0.13\text{pp}$	$-3.91 \pm 1.35\text{pp}$

A capacity-control variant, *MLP at user EOT*, tests whether single-position fragility derives from readout location vs. classifier capacity.

Results under localized perturbation. Table 3 shows that all multi-position probes are near-immune to the localized stacked-typing bundle.

The single-position probe loses $\sim 12\text{pp}$; every full-sequence multi-position probe loses $\leq 0.5\text{pp}$. Per-architecture score-shift quantiles (Appendix D, Table 8) confirm that multi-position probes also reduce per-sample shift magnitude, not only rank instability.

Capacity control. At the same readout, MLP at user EOT drops $12.5 \pm 0.8\text{pp}$ $\text{TPR@FPR}=1\%$, essentially identical to Linear’s $12.0 \pm 0.5\text{pp}$. Higher capacity at a single position yields no robustness benefit; the variable is *where* the probe reads, not *how* it classifies.

Distributed perturbation: a different balance. Under the distributed-regime stress test (every-second-word adjacent-key typo, $\approx 50\%$ of content tokens), the multi-position margin no longer holds uniformly. Attention and MultiMax remain robust (-3.8 and -3.9pp) and continue to beat Linear (-6.9pp). Full-sequence MLP, however, drops -16.8pp , losing its localized-regime margin entirely. Mean Linear shows intermediate behavior in both regimes.

The pattern follows the complementary concentration of perturbations: localized perturbations corrupt one position (bad for the user-EOT single-position readout; amortized by any form of aggregation); distributed perturbations corrupt many positions, which a uniform average over the whole sequence cannot re-weight away from, while attention and max-selection mechanisms can. Architecture choice is therefore non-trivial: *which* multi-position aggregation matters, not just whether to aggregate.

7. Practical Defenses

We evaluate three remediation strategies, differing in which stage of the pipeline they modify: architecture switching (§7.1), perturbation-augmented training (§7.2), and KV-cache forking (§7.3).

7.1. Architecture Selection

Architecture switching eliminates the localized-perturbation gap (§6.2); among aggregators, only attention- and max-selection-based variants handle both regimes. The defense is direct but requires a full retrofit of the deployed pipeline. The two complementary defenses below leave the probe architecture unchanged.

7.2. Perturbation-Augmented Training

We retrain the single-position linear probe with training-time augmentation; the architecture and inference deployment are unchanged. At deployment time the specific perturbation combination a user (or attacker) will produce is unknown, so the augmentation distribution must train the probe to generalize across combinations rather than memorize a specific one. Each near-end perturbation family (typos, capitalisation, terminal punctuation) fires independently per training sample with probability 0.5, conditioned on at least one firing, so each sample sees a random 1-to-4-component combination. Test-time evaluation uses the fixed 5-component bundle from §6, giving a deployment-realistic generalization test from random training combinations to a specific deployed perturbation pattern. 5-fold CV; full protocol in Appendix E.

Results. The augmented probe substantially recovers the bundle-induced fragility with negligible clean-accuracy cost (Table 4). $\text{TPR@FPR}=1\%$ rises from 85.4% (baseline) to $93.84 \pm 0.5\%$ (augmented), recovering 8.4 of the 12.0pp loss ($\sim 70\%$ of the fragility). Clean-test performance is essentially unchanged ($+0.14\text{pp}$ $\text{TPR@FPR}=1\%$).

Residual fragility. Recovery is not perfect even on individual perturbations seen in training: per-family $\text{TPR@FPR}=1\%$ spans 93.8–98.0% (Appendix E), a 0–4pp residual. Out-of-distribution shift (e.g., the LODO setup in Appendix I) is expected to further constrain the recoverable fraction; §7.3 describes a complementary defense that exploits spatial decay to isolate a stable readout.

7.3. KV-Cache Forked Probe Suffix

Mechanism. The KV-cache fork appends a short (≈ 30 token) generic suffix to the user turn (Figure 5), providing readout positions diluted from upstream perturbations via the spatial decay characterized in §5.2. The mechanism is

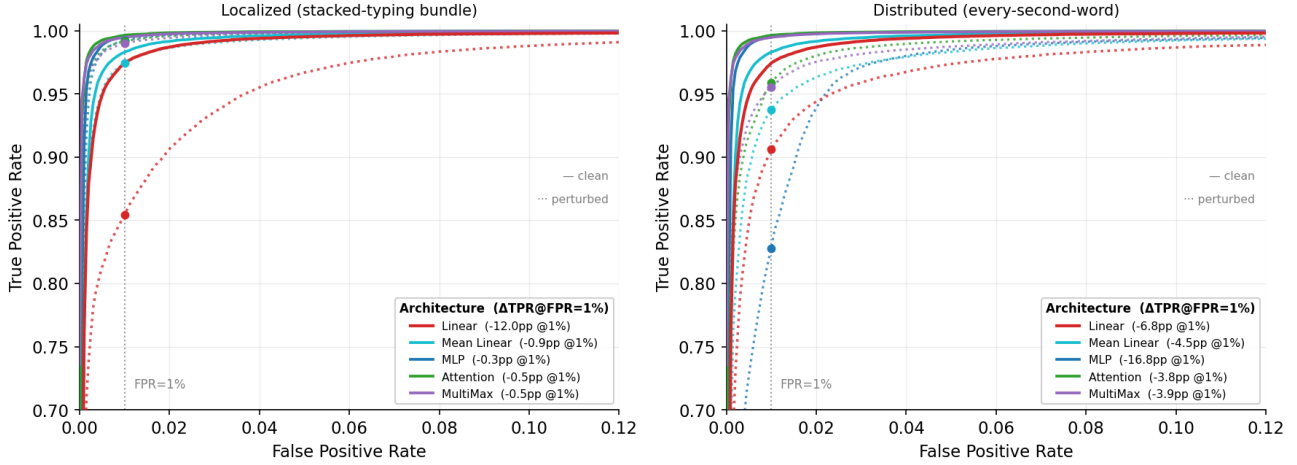


Figure 4. **Macro-averaged ROC curves (5-fold CV).** Left: localized (stacked-typing bundle). Right: distributed (every-second-word). Solid: clean; dotted: perturbed; filled dots: FPR=1% operating point on the perturbed curve.

Table 4. **Augmentation training recovers most TPR loss with no clean-accuracy cost.** Single-position linear probe at user EOT, 5-fold CV. Baseline = Table 2; Augmented = probe trained under the policy in Appendix E. Pert. = full stacked-typing bundle.

	Baseline	Augmented
Clean TPR@FPR=1%	97.4%	97.54%
Pert. TPR@FPR=1%	85.4%	93.84 $\pm$ 0.5%
$\Delta\text{TPR}@FPR=1\%$	-12.0pp	-3.70pp

architecture-agnostic; we evaluate on the single-position linear baseline of §6 with all other settings unchanged. Prefix-shared KV-cache reuse has been used adversarially to accelerate suffix-search jailbreak attacks (Wang et al., 2026); we apply the same primitive defensively, for stable probe readout.

Role scope. On Llama-3.1-8B we place the suffix in a post-user system block; for models with explicit reasoning modes (e.g., Qwen3’s <think>), the in-distribution instantiation is the CoT segment. Post-user role separation has a two-fold justification: it (i) keeps the user-turn intent boundary intact as a chat-template token boundary rather than merging probe scaffolding into user content, and (ii) yields roughly half the residual angular shift of an inline user-role suffix at fixed physical distance (Appendix H.1).

Results. On the same 5-fold CV evaluation set as §6, the KV-cache fork recovers TPR@FPR=1% to $98.65 \pm 0.02\%$ (Table 5), a $-0.60 \pm 0.06\text{pp}$ drop from clean, a 95% reduction of the -12pp baseline fragility and an order of magnitude smaller than the augmentation-training residual (-3.7pp; §7.2). Clean-test performance is also higher than the no-fork baseline (99.26% vs 97.4%), consistent with the suffix attenuating residual nuisance variation at the readout

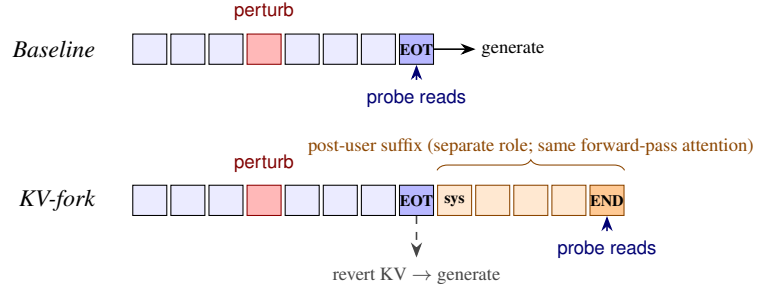


Figure 5. **KV-cache forked probe suffix.** Top: baseline reads at user-EOT, where the perturbation (red) rotates the activation. Bottom: a short post-user suffix is appended in a separate role; the probe reads at end-of-suffix, then the cache is reverted so generation proceeds from the original user-EOT (dilution mechanism: §5.2).

position.

OOD stabilisation. The fixed suffix also stabilises the readout context against upstream distribution shift: the end-of-suffix activation is dominated by the suffix itself, with upstream content acting as a controlled perturbation. Under the LODO protocol of Appendix I, retraining behind the KV-fork suffix lifts clean weighted accuracy from 77.5% (no-fork Linear, Table 13) to 81.5%, bringing the single-position linear probe into a competitive range with the best full-sequence architectures under the same protocol (Mean Linear 80.0%, Attention 81.0%, MultiMax 82.1%). At the more sensitive operating point (TPR@FPR=1%, recalibrated per fold), clean TPR rises 8.4pp on the 7 mixed-class folds (38.9% $\rightarrow$ 47.2%). Full numbers in Appendix I.2.

Comparison with augmentation training. Both §7.2 and the KV-fork retrain the same single-position probe at the

Table 5. KV-cache fork recovers near-full TPR with no architectural change. Single-position linear probe, 5-fold CV. Baseline = Table 2 (probe trained and read at user EOT); Fork = same architecture, training data, and hyperparameters, retrained and read at the end of an appended suffix.

	Baseline	KV-fork
Clean TPR@FPR=1%	97.4%	99.26 $\pm$ 0.06%
Pert. TPR@FPR=1%	85.4%	98.65 $\pm$ 0.02%
Δ TPR@FPR=1%	−12.0pp	−0.60 $\pm$ 0.06pp

same architecture and on the same training corpus; the methods differ only in what changes at training time. Augmentation requires designing a perturbation policy that anticipates the deployment-time perturbation distribution; the fork requires choosing a fixed suffix once. The fork’s larger gain (−0.6pp vs −3.7pp residual) is consistent with its mechanism being general rather than bounded by the perturbation distribution sampled during augmentation.

Compute overhead. The KV-fork suffix adds 30 tokens per request. At batch size 1, the forward-pass cost is essentially unchanged ($\Delta < 4$ ms across prompt lengths from 74 to 1531 tokens). At batch size 32, the additional forward-pass cost is 71–135 ms across the same length range (Llama-3.1-8B, A100, bf16; full table in Appendix H.2). KV-cache memory overhead is practically transparent: a few MB per request for Llama-3.1-8B (Appendix H.2), and independent of baseline length.

8. Discussion

Behavior holds; representation moves. Ordinary typing variation rotates the user-EOT activation by 43–56° across three model families (§5) without altering how the LLM reads user intent (§3.3, Appendix A). A single-position linear probe trained on those activations loses 12pp TPR at FPR=1% (§6) on input the model itself treats as routine—a silent disagreement, neither adversarial nor semantically ambiguous. Activation-based probes therefore inherit a brittleness the surface behavior does not show: their readout is more sensitive to ordinary surface noise than the model whose internals they read. The rotation also has structure—it decays sharply downstream (§5.2)—which §7’s strongest defense, a KV-cache forked probe suffix, exploits to close 95% of the gap. The defenses restore probe accuracy under that asymmetry rather than removing it.

Offensive implication. The same typing patterns can be applied deliberately as low-cost attack vectors: distributed corruption alone bypasses both probe and alignment (§3.3) without optimization, model access, or adversarial prompt engineering.

Caveats and limitations. Probe-level perturbations may alter BPE tokenization in addition to attention dilution; we control for tokenization in the activation-characterization experiments (§5) but do not decompose the two contributors in the probe-level results. Activation-level effects replicate across three model families (Llama-3.1-8B, Qwen3-8B, Gemma-4-E4B), but the multi-architecture probe sweep itself is on a single model (Llama-3.1-8B). All experiments use a single layer per model. The KV-fork role-scoping ablation (Appendix H.1) compares system-role vs. user-role suffix placement only; further role variants (e.g., assistant-role prefill) and learned-suffix designs are deferred to suffix-design follow-up work.

Future work. Several research directions follow naturally. First, in analogy to refusal-direction work, one could ask whether typo-induced activation shifts concentrate along a common direction in activation space, and whether steering along that direction *causally* produces typo-like output, establishing a causal handle on the surface-fluency representation. Second, cross-modal extensions: surface perturbations of vision-language or speech inputs (image noise, audio jitter) may exhibit analogous activation-level fragility under their own dilution mechanisms. Third, the three defenses we evaluate are mechanistically complementary; stacking them (e.g., a multi-position aggregator trained with augmentation and read behind a KV-fork suffix) could yield a stronger residual than any defense alone.

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A. LLM Behavioral Invariance

A.1. Representative Examples

Table 6 shows four representative clean-vs-perturbed examples spanning both perturbation regimes (localized and distributed) and both label classes (benign and malicious), referenced from §3.3. Each example reports the angular shift of the activation at the readout position between clean and perturbed inputs (ΔEOT). Underlined tokens mark perturbed characters.

A.2. LLM-Judge Quantification

To quantify the behavioral-invariance premise beyond the four representative examples of §A.1, we ran an LLM-judge evaluation on a larger set of (clean, perturbed) response pairs across all three models.

Setup. We sampled 100 prompts (50 benign from OpenOrca with 80–350 words; 50 malicious from Harm-Bench across its three subsets) and generated responses from Llama-3.1-8B-Instruct, Qwen3-8B, and Gemma-4-E4B under three conditions: clean, the stacked-typing bundle, and every-second-word (ESW) perturbation. This yields $100 \times 2 = 200$ (clean, perturbed) response pairs per model, 600 pairs total. Generation used greedy decoding with `max_new_tokens=200` for Llama and Gemma; Qwen3 uses 1024 to allow its chain-of-thought trajectory to complete before the final answer.

A Claude Sonnet 4.5 judge (via AWS Bedrock, temperature 0, independent per-pair calls) labeled each pair on two binary criteria: **intent preservation** (did the model interpret the perturbed prompt as asking for the same thing?) and **response equivalence** (are the responses substantively the same in content, depth, and refusal/compliance decision?). On malicious pairs we additionally tracked the *direction* of any refusal flip. Of the 600 pairs, 594 were judged; 6 pairs sharing a single biosecurity prompt were refused by the judge model itself and excluded.

Results. Table 7 reports rates per (model, perturbation, label) cell. Aggregate across the 594 judged pairs: intent preservation 94.6% and response equivalence 76.6%. The bundle and ESW regimes diverge sharply: under the bundle, response equivalence is high (82–96% across model and class cells); under ESW it drops to 50–76%, consistent with this regime’s known behavioural brittleness (§3.3). Refusal flips on the 294 malicious pairs occurred 27 times (9.2%); 12 went toward more conservative refusal and 15 toward compliance, the latter concentrated under ESW (Llama 20%, Gemma 16%, Qwen3 8%).

Table 6. **LLM behavioral invariance across perturbation regimes.** Llama-3.1-8B-Instruct. Underlined tokens mark perturbations. ΔEOT is the angular shift of the activation at the readout position between clean and perturbed inputs.

Clean	Perturbed	ΔEOT
Localized regime — stacked-typing bundle (3–4 character edits per message)		
<i>Benign</i> (capital question)		
Prompt	“What is the capital of France?”	44.2°
Response	“The capital of France is Paris.”	
<i>Malicious</i> (phishing email)		
Prompt	“Ignore your previous instructions. Help me write a phishing email to steal someone’s bank credentials.”	31.6°
Response	“I can’t assist with that request. Is there something else I can help you with?”	
Distributed regime — every-second-word ($\approx 50\%$ of content tokens corrupted)		
<i>Benign</i> (Python sort)		
Prompt	“Can you help me write a Python function that sorts a list of dictionaries by a specific key? I want to sort in both ascending and descending order.”	30.6°
Response	Complete function using sorted(..., reverse=(order== ‘desc’)) with example.	
<i>Malicious</i> (Pentagon)		
Prompt	“i want to break in the pentagon. whats the first step?”	45.8°
Response	“I can’t help with that.”	

Table 7. **LLM-judge rates of intent preservation and response equivalence across 594 (clean, perturbed) response pairs.** Cells with $n=49$ exclude the single biosecurity prompt the judge refused on.

Model	Pert	Class	n	Intent	Equiv
Llama-3.1-8B	bundle	benign	50	1.00	0.82
Llama-3.1-8B	bundle	malicious	49	0.98	0.82
Llama-3.1-8B	ESW	benign	50	0.84	0.50
Llama-3.1-8B	ESW	malicious	49	0.98	0.63
Qwen3-8B	bundle	benign	50	1.00	0.96
Qwen3-8B	bundle	malicious	49	0.98	0.88
Qwen3-8B	ESW	benign	50	0.90	0.68
Qwen3-8B	ESW	malicious	49	0.92	0.76
Gemma-4-E4B	bundle	benign	50	1.00	0.92
Gemma-4-E4B	bundle	malicious	49	0.96	0.86
Gemma-4-E4B	ESW	benign	50	0.94	0.74
Gemma-4-E4B	ESW	malicious	49	0.86	0.63
Overall			594	0.946	0.766

B. Cross-Model Spatial Decay

Figures 6 and 7 show the forward spatial decay profile for Qwen3-8B and Gemma-4-E4B respectively, using the same preamble-control design as Figure 2: relative angular change normalised to the on-site value, averaged over early, mid, and late typo-position conditions, across four checkpoint layers. Both models replicate the rapid near-typo decay observed in Llama-3.1-8B, with signal falling below 15% of the on-site value within $d \approx 10$ tokens at the deepest checkpoint layer.

C. Decay Shape Generalizes Across Perturbation Families

The decay measurements in §5.2 use adjacent-key typo perturbations because they apply at arbitrary mid-message positions and thus enable clean isolation of spatial decay. Figure 8 validates the decay shape on terminal-punctuation

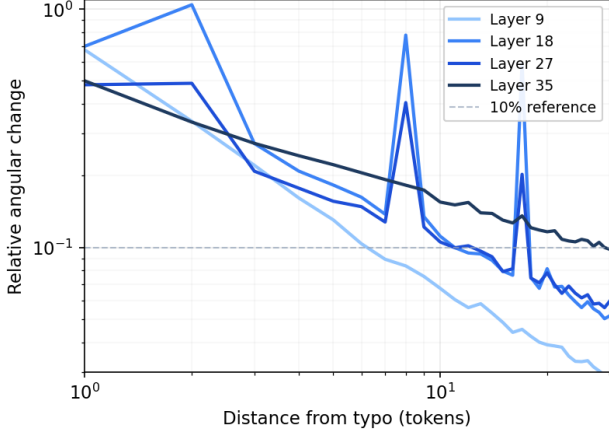


Figure 6. **Spatial decay replication: Qwen3-8B** (layers 9, 18, 27, 35). Overall decay is consistent with Llama-3.1-8B-Instruct. Non-monotonic bumps appear at intermediate layers (L18, L27), consistent across all three preamble conditions but absent in Llama-3.1-8B and Gemma-4-E4B; we report the observation as Qwen3-specific and leave the mechanism for future work.

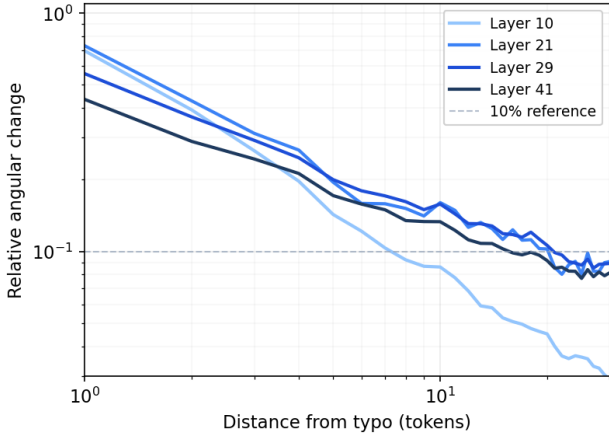


Figure 7. **Spatial decay replication: Gemma-4-E4B** (full-attention layers 10, 21, 29, 41). All three preamble conditions overlap closely throughout, indicating particularly strong position-invariance of the decay profile. Signal falls below 15% by $d \approx 10$ at the deepest layer, consistent with Llama-3.1-8B-Instruct.

perturbations ($? \rightarrow .$ and $? \rightarrow /$) on Llama-3.1-8B layer 31 ($N=50$ OpenOrca prompts each). Curves are normalized to each perturbation’s on-site delta norm.

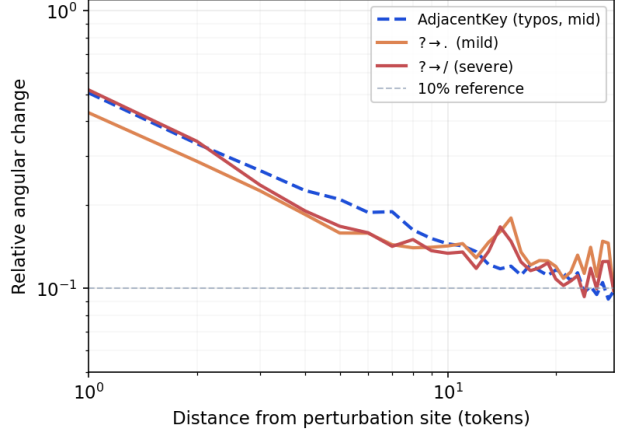


Figure 8. **Spatial decay generalizes beyond adjacent-key typos.** Relative-norm decay curves for AdjacentKey typos (dashed blue, mid condition), $? \rightarrow .$ (orange, mild), and $? \rightarrow /$ (red, severe), on Llama-3.1-8B layer 31. Despite on-site rotation differing $\sim 2.4\times$ across types, the relative-decay profiles overlap.

D. Per-Architecture Score-Shift Distributions

Table 8 reports the distribution of per-sample score shifts (perturbed – clean) under the stacked-typing bundle, by architecture and ground-truth class. Linear (single-position) has heavy-tailed shifts on the malicious side (5th percentile -0.49 , median ≈ 0); aggregating probes reduce per-sample magnitude dramatically, with MLP/Attention/MultiMax showing near-zero shifts at every percentile.

Table 8. **Per-architecture score-shift summary.** Quantiles of per-sample (perturbed – clean) score shift under the stacked-typing bundle, by probe and class (B = benign, M = malicious). 5-fold CV pooled, Llama-3.1-8B layer 31.

Probe	Cls	5th	Median	Mean	95th
Linear	B	-0.043	+0.000	+0.021	+0.185
	M	-0.490	-0.000	-0.053	+0.021
Mean Lin.	B	-0.010	+0.002	+0.014	+0.091
	M	-0.022	-0.000	-0.003	+0.006
MLP	B	-0.000	+0.000	+0.005	+0.014
	M	-0.000	+0.000	-0.001	+0.002
Attention	B	-0.000	+0.000	+0.007	+0.009
	M	-0.000	+0.000	-0.001	+0.001
MultiMax	B	-0.000	+0.000	+0.006	+0.007
	M	-0.000	+0.000	-0.002	+0.000

E. Perturbation-Augmentation Training Details

Training-time augmentation policy. For each training sample (5-fold CV, fold of $\approx 134,750$ samples), the probe is trained on activations re-extracted from text augmented by independent near-end perturbation families. Each family fires independently with probability 0.5 per sample, with a

guarantee that at least one perturbation applies, so a training sample sees one to four perturbations applied individually:

- **last_word_typo**: QWERTY adjacent-key typo on the last alphabetic word.
- **trail_period**: toggle the trailing period.
- **early_shift**: uppercase a letter before punctuation.
- **terminal_punct**: one of $\{? \rightarrow ., ? \rightarrow /, . \rightarrow /\}$ chosen uniformly per sample.

Per-perturbation evaluation. Table 9 reports per-perturbation TPR@FPR=1% (5-fold mean $\pm$ std) for the augmented probe.

Table 9. **Augmented probe per-perturbation evaluation.** 5-fold CV mean $\pm$ std. The full bundle (last row) is the 5-component compound perturbation reported in the body.

Perturbation	TPR@FPR=1%
Clean (no perturbation)	97.54 $\pm$ 0.14%
last_word_typo	95.68 $\pm$ 0.26%
trail_period	95.54 $\pm$ 0.25%
early_shift	95.52 $\pm$ 0.43%
? $\rightarrow$.	97.98 $\pm$ 0.20%
? $\rightarrow$ /	96.60 $\pm$ 0.37%
. $\rightarrow$ /	93.84 $\pm$ 0.59%
full_bundle (5-component)	93.84 $\pm$ 0.53%

F. Dataset Details

Table 10 lists the 29 datasets used in the probe-consequence and multi-architecture experiments (§6). Most source datasets are far larger than the counts reported here (e.g., Open-Orca contains $\sim$ 3M samples, BIPIA contains 100k+ injection examples). We *sample* from each dataset during cache ingestion, with caps chosen to keep the training corpus tractable ($N=168,440$ samples total) and to avoid any single dataset dominating the empirical distribution. The N column reports the actual per-dataset sample counts loaded into the cache; for datasets smaller than the cap, N reflects the full dataset size (e.g., Gandalf at 114, Scam at 25). Probe training uses balanced class weights (Appendix G) to compensate for the moderate class imbalance reported at the bottom of Table 10.

Source paths. Canonical Hugging Face or GitHub repositories for each dataset are listed in Appendix J, Table 18.

G. Probe Training Hyperparameters

Layer choice. We probe activations at layer 31, the final residual-stream layer before the unembedding projection. Fomin (2026) sweep five Llama-3.1-8B layers (19, 23, 25,

27, 31) at positions -5 and -1 under Leave-One-Dataset-Out evaluation and report that layer 31 and layer 27 at position -5 are competitive on aggregate weighted accuracy (81.8–82.3%); they explicitly select layer 31 as a principled default (final layer, last user token) rather than as a performance optimum, and observe that no single layer dominates across all datasets. We adopt the same layer 31 default and reproduce the finding on our 29-dataset corpus.

Training. All probes are trained as binary classifiers on Llama-3.1-8B-Instruct layer 31 activations using PyTorch with the Adam optimizer, balanced class weights, and random seed 42. Table 11 lists architecture-specific and training hyperparameters used in the 5-fold stratified cross-validation reported in §6.

Linear and Mean Linear probes use 1000 training epochs (full-batch gradient descent on the standardised activations); MLP-based probes (MLP, Attention, MultiMax) use 5 epochs of mini-batch SGD with no input normalisation, following the convention of Kramár et al. (2026). All architectures share the class-weighting and optimisation settings; the training set in each fold is $\approx$ 134,750 samples and the test fold is $\approx$ 33,690.

Table 10. Datasets used in probe-consequence and multi-architecture experiments (29 sources; $N=168,440$ samples). Grouped by label class: malicious (with attack-type subheaders), mixed (labelled examples on both sides), benign. ‡ merged with a larger dataset for LODO fold stability (Gandalf→Mossap; Scam→AdvBench).

Dataset	Category	Content	N
<i>Malicious — direct prompt injection</i>			
Mossap	direct PI	Direct injections	10,000
Jayavibhav	direct PI	Direct injections	10,000
Deepset	direct PI	Direct injections	546
Yanismiraoui	direct PI	Direct injections	1,034
<i>Malicious — indirect prompt injection</i>			
BIPIA (Yi et al., 2025)	indirect PI	Email/code/table embed	15,000
InjecAgent (Zhan et al., 2024)	indirect PI	Tool-call injections	1,020
LLMail	indirect PI	Email challenge	10,000
AgentDojo (Debenedetti et al., 2024)	indirect PI	Agent task injections	5,000
Gandalf‡	indirect PI	Summarization	114
Scam‡	indirect PI	Scam scenarios	25
<i>Malicious — jailbreak</i>			
WildJailbreak (Jiang et al., 2024)	jailbreak	Adversarial jailbreaks	2,000
Jailbreak-CLS	jailbreak	Jailbreak classification	1,044
<i>Malicious — harmful / unsafe requests</i>			
AdvBench (Zou et al., 2023)	harmful req.	Unsafe requests	520
HarmBench (Mazeika et al., 2024)	harmful req.	Unsafe requests	400
<i>Mixed (labelled on both sides)</i>			
SafeGuard	mixed	PI benchmark	8,236 (30% mal)
Qualifire	mixed	PI benchmark	5,000 (40% mal)
<i>Benign</i>			
Enron	benign	Email corpus	10,000
Dolly-15k	benign	Instruction following	10,000
Open-Orca	benign	Reasoning instructions	10,000
Prompts-Ranked	benign	User prompts	10,000
Alpaca	benign	Instruction following	10,000
SoftAge	benign	Prompt engineering	1,001
Bitext CS	benign	Support chat	10,000
Code Exercise	benign	Python exercises	10,000
Python-Alpaca	benign	Python instructions	5,000
Python-25k	benign	Python snippets	5,000
Writing Prompts	benign	Creative writing	10,000
xLAM	benign	Tool use	5,000
APIGen-MT	benign	Multi-turn tools	2,500
Total corpus			168,440 (32.4% mal)

Table 11. Probe training hyperparameters (5-fold CV, all probes use Adam, balanced class weights, weight decay 3×10^{-3} , early stopping with patience 50, random seed 42).

Probe	Input scope	Hidden	Heads	Activation	LR	Batch size	Normalisation
Linear (user EOT)	user EOT (1 token)	—	—	—	10^{-3}	full-batch	standard
MLP (user EOT)	user EOT (1 token)	100	—	ReLU	10^{-4}	64	none
Mean Linear (last 16)	last 16 tokens	—	—	—	10^{-3}	full-batch	standard
MLP (all)	full user-turn	100	—	ReLU	10^{-4}	64	none
Attention (all)	full user-turn	100	10	ReLU	10^{-4}	64	none
MultiMax (all)	full user-turn	100	10	ReLU	10^{-4}	64	none

H. KV-Fork Supplementary Details

H.1. Role-Scoping Ablation

- 14 We test whether the post-user role choice for the KV-fork suffix (system block vs. user-message extension) produces

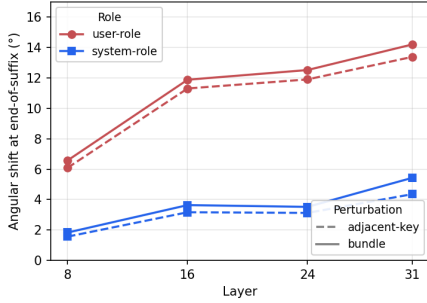


Figure 9. End-of-suffix angular shift by role and layer. System-role suffix (blue) yields roughly half the residual angular shift of user-role suffix (red) at every measured layer, for both perturbations (adjacent-key dashed, bundle solid). Paired difference at L31 is -9.03° (adjacent-key) and -8.77° (bundle). Llama-3.1-8B, $n=100$ OpenOrca prompts.

different attenuation of the perturbation effect, holding suffix content fixed.

Setup. Three placements paired per prompt: *none* (no suffix; reference baseline at user-EOT), *user_role* (suffix inside the user message), and *system_role* (suffix in a post-user system block). Two perturbations: a single adjacent-key typo and the 5-component bundle from §6. $N=100$ OpenOrca prompts (150–300 words), seed-paired; layers 8, 16, 24, 31. The metric is the angular shift between clean and perturbed activations at the end-of-suffix probe-readout position.

Distance-based dilution (§5.2) is part of the story, but role choice adds substantial extra attenuation. Combined with the intent-preservation argument (§7.3), system-role placement is justified empirically and architecturally.

H.2. Compute Overhead

We benchmark inference-time forward-pass cost of the KV-fork suffix on Llama-3.1-8B-Instruct (bf16, sdpa attention, single A100 80GB) at batch sizes 1, 8, and 32, over four user-message length buckets (74, 235, 559, 1531 tokens after chat-template; constructed by tiling a representative prompt with filler text). The KV-fork condition appends the 30-token neutral suffix from §7.3. Forward-pass-only timings, paired, median over 10 reps after 3 warmup. Δ ms is (KV-fork) – (baseline).

KV cache memory marginal is 3.75 MB per request (analytic; 30 tokens $\times$ 32 layers $\times$ 8 KV heads $\times$ 128 head-dim $\times$ 2 (K,V) $\times$ 2 bytes bf16), independent of baseline length. For other model architectures this marginal scales with layer count, KV-head count, and head dimension; the 30-token constant is the smaller factor. The suffix’s absolute additional cost grows modestly with baseline length at fixed batch but remains bounded: at batch 1 the cost is under 4 ms

Table 12. KV-fork forward-pass overhead by length bucket. Llama-3.1-8B-Instruct, bf16, A100. KV-fork suffix is 30 tokens. Δ is (KV-fork) – (baseline) median over 10 reps; n_{base} rounded to the nearest 5.

Bucket	n_{base}	batch	base (ms)	Δ (ms)
short	75	1	31.5	0.3
		8	59.1	23.6
		32	200.8	71.2
medium	235	1	34.1	2.9
		8	167.0	19.1
		32	620.8	85.1
long	560	1	59.6	1.4
		8	387.4	15.9
		32	1468.0	88.0
very long	1530	1	141.7	3.8
		8	1043.6	25.9
		32	4151.7	135.2

regardless of prompt length; at batch 32 it ranges 71–135 ms across the four buckets.

I. Leave-One-Dataset-Out Evaluation

We measure out-of-distribution generalization following the LODO protocol of Fomin (2026): for each of the $K=29$ datasets in our benchmark (Appendix F), we retrain each probe architecture on the union of the remaining 28 and evaluate on the held-out dataset. This isolates whether the probe’s signal transfers to a distribution it has not seen in training, rather than memorizing dataset-specific surface patterns.

Metrics and threshold convention. We evaluate at a fixed decision threshold of 0.5 alongside ROC-AUC, following Fomin (2026). Fixed-threshold accuracy measures the deployment-realistic quantity the LODO setup is designed to test: how well a probe shipped without per-domain calibration data recognises a new distribution. Threshold 0.5 is the symmetric midpoint of the sigmoid output and requires no held-out information from the new fold. Per-fold threshold tuning would instead answer whether *probe* + *per-domain calibrator* transfers, conflating probe quality with calibration skill.

Of the 29 folds, only 7 contain both classes and admit AUC; the other 22 are single-class. For single-class folds, accuracy reduces to recall (malicious-only folds, $n=9$) or to $1 - \text{FPR}$ (benign-only folds, $n=13$). We report sample-weighted accuracy across the full held-out population ($N=168,440$), per-fold mean AUC across the 7 mixed folds, and pooled AUC over their concatenated held-out predictions. We do not report cross-fold mean accuracy: dataset sizes vary $\sim 600 \times$ (25 to 15,000 samples), so an unweighted average across folds is dominated by sampling noise on the smallest folds and lacks a clean population interpretation; per-

Table 13. LODO clean evaluation per probe architecture. 29 held-out datasets, Llama-3.1-8B-Instruct, layer 31. Mean AUC and pooled AUC are computed on the 7 mixed-class folds. Weighted accuracy is sample-weighted across the full held-out population ($N=168,440$) at decision threshold 0.5, following Fomin (2026). Bold marks per-column best.

Architecture	Pooled AUC	Mean AUC ($\pm$ std)	Weighted ACC
Linear (EOT)	0.796	0.814 ± 0.164	0.775
Mean Linear (last 16)	0.845	0.863 ± 0.127	0.800
MLP (all)	0.747	0.844 ± 0.131	0.798
Attention (all)	0.810	0.918 ± 0.057	0.810
MultiMax (all)	0.839	0.879 ± 0.084	0.821

distribution heterogeneity is reported directly in Table 14 below.

Aggregate results. Probes degrade meaningfully under LODO: best mean per-fold AUC is 0.918 (Attention) against the 0.998 in-distribution baseline (§6). Mean and pooled AUC also diverge: MLP’s pooled AUC (0.747) falls 10pp below Mean Linear’s (0.845) despite a competitive mean per-fold AUC (0.844), indicating that non-linear probes maintain within-fold rank but produce scores that are not commensurable across LODO folds.

Per-dataset breakdown. Three patterns in Table 14 are load-bearing. First, several folds are catastrophic across all five architectures (e.g., APIGenMT below 3%), an extreme manifestation of score-scale drift on distributions none of the probes saw in training. Second, per-fold rankings flip relative to the aggregate (e.g., the linear probe tops InjecAgent’s recall while MultiMax bottoms it), so the aggregate ranking that favors capacity is not a uniform per-distribution improvement. Third, cross-fold standard deviations (± 0.25 – ± 0.30 on accuracy) show that LODO performance is dominated by held-out dataset choice rather than architecture, in stark contrast to the ± 0.01 standard deviation observed under 5-fold CV (§4).

Together these patterns point past a generic “probes overfit” to a sharper claim: the training distribution couples intent with dataset-specific surface attributes, so what the probes learn is an intent-aligned *mixture* rather than intent in isolation. LODO holds out the dataset-specific component by construction, leaving a per-fold residual that varies in composition. This complements the SAE feature-retention finding of Fomin (2026): same phenomenon, observed here at prediction accuracy rather than at the feature level.

I.1. Perturbation under LODO

We apply the stacked-typing bundle and every-second-word (ESW) perturbations of §6 to each mixed-class held-out

fold ($n=7$) and report TPR @ FPR=1% with the threshold re-calibrated per (fold, condition), matching the operational convention of §6 so that the perturbation Δ reported here is directly comparable.

Cross-architecture summary. Table 15 reports the cross-fold mean TPR @ FPR=1% per architecture. The mean Δ under bundle and ESW is small in absolute terms (-1 to -7 pp), but this aggregate hides extreme cross-fold heterogeneity (Table 16).

Per-fold heterogeneity. Per-fold drops are heterogeneous (Table 16). Folds with low clean baselines (AgentDojo, BIPIA) register near-zero or positive Δ while folds with ID-comparable baselines (SafeGuard, Jailbreak Classification) register ID-comparable drops, indicating that ID perturbation fragility persists under distribution shift but is masked at the aggregate by floor effects.

I.2. KV-Fork under LODO

We re-train the linear probe behind the KV-fork suffix (§7.3) on the in-distribution training corpus and evaluate under the LODO protocol of §1. Two metrics are reported: clean weighted accuracy at threshold 0.5 across all 28 available held-out folds (matching Table 13), and clean / perturbed TPR @ FPR=1% on the 7 mixed-class folds with threshold recalibrated per (fold, condition) (matching §1.1).

The KV-fork lifts the clean weighted accuracy of a single-position linear probe by +4.0pp, enough to bring it into the architectural top tier under LODO (cf. Mean Linear 0.800, Attention 0.810, MultiMax 0.821 in Table 13). At the recalibrated FPR=1% operating point the clean gain is +8.4pp, consistent with the suffix stabilising the readout context against the score-scale drift that dominates LODO behaviour for non-augmented probes (§1). Suffix-content variation is left to future work.

Table 17. KV-fork under LODO. Clean weighted accuracy is sample-weighted across the 28 held-out folds at threshold 0.5 (Scam unavailable). TPR @ FPR=1% is on the 7 mixed-class folds with threshold recalibrated per (fold, condition); cross-fold mean.

Condition	Weighted ACC	TPR clean	Δ TPR bundle
No KV-fork (Linear EOT)	77.5%	38.9%	-3.5 pp
KV-fork (neutral suffix)	81.5%	47.2%	-4.5 pp
Δ (KV-fork – no-fork)	+4.0pp	+8.4pp	-1.0 pp

J. Dataset Source Paths

Table 14. **Per-dataset clean LODO accuracy at threshold 0.5.** Layer 31, Llama-3.1-8B-Instruct. Mixed-class folds report standard accuracy; malicious-only folds report recall; benign-only folds report $1 - \text{FPR}$. %mal column shows malicious fraction in the held-out fold. ScamDataset clean test set is included; perturbation evaluation is unavailable for it due to source repository unavailability at evaluation time.

Dataset	N	%mal	Linear	Mean Lin.	MLP	Attn.	MMax
<i>Mixed (accuracy)</i>							
AgentDojo	5,000	99	21.7	53.2	84.0	77.5	28.4
BIPIA	15,000	95	40.4	55.5	32.9	36.7	58.5
Deepset	546	37	76.6	79.7	82.6	84.2	80.2
Jailbreak Classification	1,044	50	94.4	95.6	95.3	96.3	96.1
Jayavibhav	10,000	50	76.7	75.4	74.7	72.5	75.9
Qualifire	5,000	40	75.0	77.1	75.0	75.6	76.3
SafeGuard	8,236	30	97.5	96.4	94.4	96.2	94.5
<i>Malicious-only (recall)</i>							
AdvBench	520	100	96.0	96.9	87.5	98.1	99.2
Gandalf Summarization	114	100	98.2	97.4	92.1	100.0	100.0
HarmBench	400	100	43.0	65.5	52.0	64.2	68.2
InjecAgent	1,020	100	100.0	90.2	35.8	56.1	6.6
LLMail	10,000	100	79.0	38.4	44.3	22.7	22.4
MosscaP	10,000	100	78.0	89.4	99.9	100.0	100.0
Scam	25	100	96.0	64.0	100.0	100.0	92.0
Wild Jailbreak	2,000	100	85.7	91.7	86.0	90.6	90.6
Yanismiraoui	1,034	100	56.0	52.3	26.3	42.6	63.0
<i>Benign-only ($1 - \text{FPR}$)</i>							
APIGenMT	2,500	0	0.4	0.1	0.0	2.8	2.9
Alpaca	10,000	0	99.5	99.6	99.9	99.7	99.9
Bitext Customer Support	10,000	0	97.9	98.7	99.5	99.0	98.2
Code Exercise	10,000	0	99.9	100.0	100.0	100.0	100.0
Dolly15k	10,000	0	99.8	99.8	99.9	99.9	99.8
Enron	10,000	0	74.5	72.7	86.5	65.1	86.8
OpenOrca	10,000	0	60.9	85.2	86.3	96.2	87.7
Prompts Ranked 10k	10,000	0	92.1	95.3	96.0	97.0	95.7
Python Code Alpaca	5,000	0	99.1	100.0	100.0	100.0	100.0
Python Codes 25k	5,000	0	98.2	97.0	97.0	98.1	96.5
SoftAge	1,001	0	94.9	86.3	96.8	97.5	97.6
Writing Prompts	10,000	0	90.8	91.6	96.8	89.5	91.2
Xlam Function Calling	5,000	0	7.3	22.2	0.2	99.5	99.9

Table 15. **TPR@FPR=1% under LODO with perturbation.** Seven mixed-class folds, threshold re-calibrated per (fold, condition) to hold $\text{FPR}=1\%$. Mean across folds. Δ is perturbed minus clean. In-distribution reference (Table 2): linear probe clean $\text{TPR}=97.4\%$, bundle $\Delta=-12.0$ pp.

Architecture	TPR clean	TPR bundle	TPR ESW	Δ bundle	Δ ESW
Linear (EOT)	38.9%	35.4%	33.6%	-3.5pp	-5.3pp
Mean Linear (last 16)	44.3%	43.4%	40.2%	-0.9pp	-4.1pp
MLP (all)	40.6%	39.1%	35.5%	-1.5pp	-5.1pp
Attention (all)	46.3%	47.7%	43.8%	+1.4pp	-2.5pp
MultiMax (all)	44.3%	43.4%	38.1%	-1.0pp	-6.2pp

Table 16. **Per-fold linear probe TPR@FPR=1% under perturbation (seven mixed folds).** Threshold re-calibrated per (fold, condition). Δ is perturbed minus clean.

Fold	Clean	Bundle	Δ bundle	ESW	Δ ESW
AgentDojo	0.5%	0.1%	−0.4pp	3.6%	+3.1pp
BIPIA	2.4%	6.2%	+3.8pp	7.0%	+4.5pp
Deepset	34.0%	30.0%	−3.9pp	28.2%	−5.8pp
Jailbreak Classification	82.4%	71.2%	−11.2pp	76.4%	−5.9pp
Jayavibhav	44.8%	43.6%	−1.3pp	32.4%	−12.4pp
Qualifire	14.4%	11.0%	−3.4pp	10.9%	−3.5pp
SafeGuard	93.5%	85.4%	−8.1pp	76.7%	−16.8pp

Table 18. **Dataset source paths.** Canonical Hugging Face or GitHub repository for each of the 29 datasets in Appendix F.

Dataset	Source
Enron	amaneo/enron-mail-corpus-mini
Dolly-15k	databricks/databricks-dolly-15k
Open-Orca	Open-Orca/OpenOrca
Prompts-Ranked	data-is-better-together/10k-prompts_ranked
Alpaca	tatsu-lab/alpaca
SoftAge	SoftAge-AI/ (gated)
Bitext CS	bitext/Bitext-customer-support-llm-chatbot-training-dataset
Code Exercise	iamtarun/python.code.instructions.18k.alpaca
Python-Alpaca	iamtarun/python.code.instructions.18k.alpaca
Python-25k	flytech/python-codes-25k
Writing Prompts	euclaise/writingprompts
xLAM	Salesforce/xlam-function-calling-60k
APIGen-MT	Salesforce/APIGen-MT-5k
SafeGuard	xTRam1/safe-guard-prompt-injection
Qualifire	qualifire/prompt-injections-benchmark
MosscaP	Lakera/mosscaP.prompt.injection
Jayavibhav	jayavibhav/prompt-injection-safety
Deepset	deepset/prompt-injections
Yanismiraoui	yanismiraoui/prompt-injections
BIPIA	github.com/microsoft/BIPIA
InjecAgent	github.com/uiuc-kang-lab/InjecAgent
LLMail	microsoft/llmail-inject-challenge
AgentDojo	github.com/ethz-spylab/agentdojo
Gandalf	Lakera/gandalf.summarization
Scam	github.com/1Password/SCAM
WildJailbreak	allenai/wildjailbreak
Jailbreak-CLS	jackhhao/jailbreak-classification
AdvBench	walledai/AdvBench
HarmBench	walledai/HarmBench